WORKED SOLUTION

STEP I 2017, Question 4

Pure • Quadratics and Geometric Series

STEP I 2017, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q4

The key idea

Everything depends on how many values of r in the open interval $(-1,1)$ give a particular value of p (or q). That is a question about a parabola restricted to $-1 < r < 1$: a horizontal line meets it once, twice or not at all, and the sketch tells us which. Once we know r, the sum is fixed, and because the sum is a one-to-one function of r, two values of r always give two different sums. To find the equation satisfied by the sums, we write r in terms of the sum and substitute.

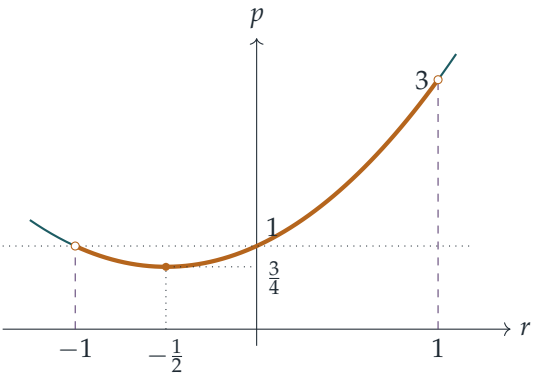
Solution

Part (i)

Completing the square,

$$1 + r + r^2 = \left(r + \tfrac{1}{2}\right)^2 + \tfrac{3}{4},$$

so the graph is a parabola with its minimum point at $\left(-\tfrac{1}{2}, \tfrac{3}{4}\right)$. It passes through $(-1,1)$, $(0,1)$ and $(1,3)$.



On $-1 < r < 1$ (the thick arc) the function decreases from 1 (not attained, since $r = -1$ is excluded) to its minimum $\frac{3}{4}$ at $r = -\frac{1}{2}$, then increases towards 3 (again not attained, since $r = 1$ is excluded). Hence

$$\tfrac{3}{4} \leq p < 3.$$

When $1 < p < 3$. The values of r giving this p are the roots of

$$r^2 + r + (1 - p) = 0.$$

The product of the roots is $1 - p < 0$, so there is one positive root and one negative root. Their sum is -1, so the negative root equals -1 minus the positive root, which is less than -1 and therefore not allowed. The positive root is

$$r = \frac{-1 + \sqrt{4p - 3}}{2},$$

and since $1 < 4p - 3 < 9$ we have $0 < r < 1$, so it is allowed. (On the graph: the line at height $p \in (1, 3)$ meets the thick arc only on the rising part $0 < r < 1$.) So p determines r uniquely, and hence $S = 1/(1 - r)$ uniquely.

When $\frac{3}{4} < p < 1$. Now the discriminant $1 - 4(1 - p) = 4p - 3$ is positive, so the quadratic has two distinct real roots. Their product $1 - p$ is positive and their sum is -1, so both are negative. Each root equals -1 minus the other, and the other is negative, so each root is greater than -1. Both roots therefore lie in $(-1, 0)$ and both are allowed.

> **Ask yourself** Two values of r, fine; but why must they give two *different* values of S?

Because $S = \dfrac{1}{1 - r}$ can be inverted: $r = 1 - \dfrac{1}{S}$. Different values of r therefore give different values of S, so there are exactly two possible values of S.

To find the equation they satisfy, substitute $r = 1 - u$ with $u = 1/S$ into $r^2 + r + 1 - p = 0$:

$$(1 - u)^2 + (1 - u) + 1 - p = 0$$
$$u^2 - 3u + (3 - p) = 0.$$

Putting $u = 1/S$ and multiplying by S^2 (which is non-zero) gives

$$(3 - p)S^2 - 3S + 1 = 0,$$

as required. Both values of S satisfy it, since both come from allowed roots r.

Part (ii)

Completing the square again,

$$q = 1 + 2r + 3r^2 = 3\left(r + \tfrac{1}{3}\right)^2 + \tfrac{2}{3}.$$

The minimum is $\frac{2}{3}$ at $r = -\frac{1}{3}$; at $r = -1$ the value is 2, and at $r = 1$ it is 6. So on $-1 < r < 1$:

- on the falling part, $-1 < r \leq -\frac{1}{3}$, q takes each value in $\left[\frac{2}{3}, 2\right)$ exactly once;
- on the rising part, $-\frac{1}{3} \leq r < 1$, q takes each value in $\left[\frac{2}{3}, 6\right)$ exactly once.

The two parts share only the point $r = -\frac{1}{3}$. Hence a value of q comes from

- exactly one r if $q = \frac{2}{3}$ (namely $r = -\frac{1}{3}$) or $2 \leq q < 6$ (rising part only);
- exactly two values of r if $\frac{2}{3} < q < 2$ (one on each part);
- no r at all otherwise.

For instance, at $q = 2$ the equation $3r^2 + 2r - 1 = 0$ factorises as $(3r - 1)(r + 1) = 0$, and the root $r = -1$ is excluded, leaving only $r = \frac{1}{3}$.

As before, T determines r: since $1 - r > 0$, we have $1 - r = 1/\sqrt{T}$. So different r give different T. Therefore

$$T \text{ is determined uniquely when } q = \tfrac{2}{3} \text{ or } 2 \leq q < 6,$$

and

$$T \text{ has two possible values when } \tfrac{2}{3} < q < 2.$$

The quadratic. Write $r = 1 - u$, where $u = 1 - r = 1/\sqrt{T} > 0$. Then

$$q = 1 + 2(1 - u) + 3(1 - u)^2 = 3u^2 - 8u + 6.$$

Since $u^2 = 1/T$, this reads $\dfrac{3}{T} + 6 - q = \dfrac{8}{\sqrt{T}}$. Multiplying by T,

$$(6 - q)T + 3 = 8\sqrt{T}.$$

This is already a quadratic in $\sqrt{T}$. To get one in T itself, square both sides:

$$(6 - q)^2 T^2 + 6(6 - q)T + 9 = 64T,$$

that is,

$$(6 - q)^2\, T^2 - 2(3q + 14)\, T + 9 = 0.$$

Both possible values of T satisfy this. Squaring can't have brought in a stray root here. With $q < 6$ and $T > 0$, the left-hand side $(6 - q)T + 3$ is positive, so the alternative $(6 - q)T + 3 = -8\sqrt{T}$ has no solutions with $T > 0$. Also, the quadratic in T has positive product $9/(6 - q)^2$ and positive sum, so its roots are both positive. So its two roots are exactly the two values of T.

Checks

- Take $p = \frac{7}{8}$. Then $r^2 + r + \frac{1}{8} = 0$ gives $r = \frac{-2 \pm \sqrt{2}}{4}$, both in $(-1, 0)$. The corresponding sums $S \approx 0.873$ and $S \approx 0.540$ satisfy $\frac{17}{8}S^2 - 3S + 1 = 0$, whose roots have sum $\frac{24}{17} \approx 1.412$ and product $\frac{8}{17}$.

- Take $q = 1$. Then $3r^2 + 2r = 0$ gives $r = 0$ or $r = -\frac{2}{3}$, so $T = 1$ or $T = \frac{9}{25}$. The quadratic becomes $25T^2 - 34T + 9 = (T-1)(25T-9)$, as it should.
- At the boundary $q = \frac{2}{3}$ the quadratic is $\frac{256}{9}T^2 - 32T + 9 = 0$, with discriminant $1024 - 1024 = 0$. It has a double root $T = \frac{9}{16} = 1/(1+\frac{1}{3})^2$, matching the single value $r = -\frac{1}{3}$.
- Thousands of random values of p and q in each range were checked by computer: the root counts and both quadratics agree.

Common slips

- Forgetting that $r = \pm 1$ are excluded, which wrongly puts $p = 1$ in the "two values" case and $q = 2$ out of the "unique" set.
- Missing the isolated value $q = \frac{2}{3}$ (the vertex), where T is also unique.
- Squaring without checking that no extra roots have been introduced.

Beyond STEP

The question is really about when a function is invertible: the map $r \mapsto p$ is one-to-one on part of its domain and two-to-one on the rest, and the answer depends on the domain, not just on the formula. The series $\sum r^n$ and $\sum nr^{n-1}$ are a power series and its term-by-term derivative. Why differentiating term by term is allowed inside the radius of convergence is a standard first-year analysis result. Abbott's *Understanding Analysis* treats power series and uniform convergence carefully and readably.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.